



OPERATING SYSTEM DESIGN

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Department of Computer Applications

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Storage Management & System Security

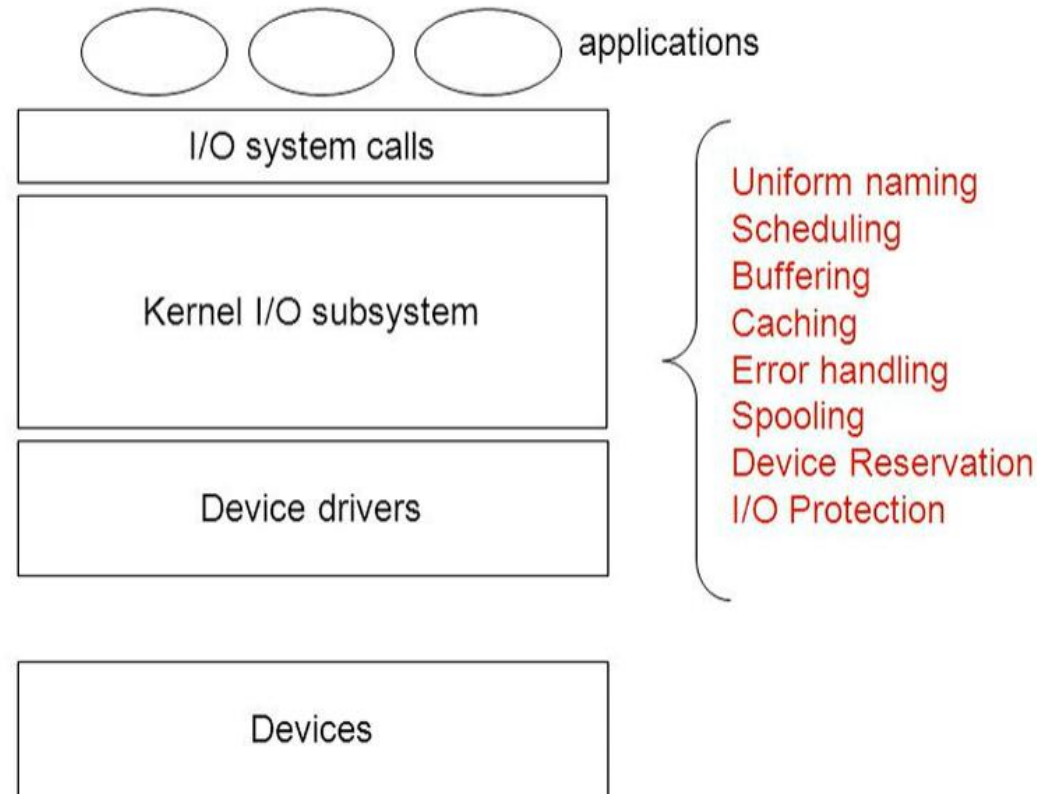
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- Kernel provides many service related to IO
- Some of the kernel services: Scheduling, Caching, spooling, device registration, error handling
- These services are built on the hardware as Kernel Subsystem which can be comprised in the device driver infrastructure
- Kernel IO subsystems are integral component of OS responsible for managing IO operations between user-space and devices

Components of Kernel I/O Subsystems

- I/O Scheduling
- Buffering
- Caching
- Spooling
- Device reservation
- Error Handling



IO Scheduling

Purpose: Optimizes disk I/O performance by prioritizing and organizing I/O requests

Algorithms: CFQ (Completely Fair Queuing), Deadline, NOOP, etc

Considerations: Latency, throughput, fairness, and workload characteristics

Buffering

Purpose: Reduces the number of I/O operations by temporarily storing data in memory

Types: Full buffering, line buffering, No buffering

Mechanism: Buffer allocation, read-ahead buffering, write-behind caching

Caching

Purpose: Stores frequently accessed data in memory to accelerate access and reduce disk I/O latency.

Types: Page cache, file system cache, disk cache.

Mechanism: LRU (Least Recently Used) algorithm, caching policies

Spooling

Purpose: Manages the simultaneous output of multiple processes or devices to a single destination.

Use Cases: Print spooling, network spooling.

Mechanism: Queuing, scheduling, and prioritizing output tasks.

Device Reservation

Purpose: Coordinates access to shared devices among multiple processes or users.

Mechanism: Locking mechanisms, access control, reservation policies.

Use Cases: Shared printers, network resources.

Error Handling

Purpose: Detects and manages errors occurring during I/O operations to ensure system integrity

Mechanism: Error codes, error recovery, fault tolerance

Use Cases: Disk errors, network errors, device failures



THANK YOU

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